

JOYKIRAT SINGH

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EDUCATION

University of North Carolina at Chapel Hill

Ph.D. in Computer Science (Advised by Prof. [Mohit Bansal](#))

Chapel Hill, NC

2025 – Present

Indraprastha Institute of Information Technology, Delhi

B.Tech in Computer Science and Design (Silver medalist); GPA: 9.24/10.0

New Delhi, India

2019 – 2023

RESEARCH INTERESTS

My research focuses on building large language models that can reason more efficiently and in an interpretable manner. Additionally, I am interested in agentic integrations that enable more adaptive and goal-directed behavior.

PUBLICATIONS

Think Right: Learning to Mitigate Under-Over Thinking via Adaptive, Attentive Compression

Joykirat Singh, Justin Chih-Yao Chen, Archiki Prasad, Elias Stengel-Eskin, Akshay Nambi, Mohit Bansal (UnderReview ICLR2026) [\[PDF\]](#)

Self-Evolved Preference Optimization for Enhancing Mathematical Reasoning in Small Language Models

Joykirat Singh, Tanmoy Chakraborty, and Akshay Nambi (Under Review AAAI 2026) [\[PDF\]](#)

Agentic Reasoning and Tool Integration for LLMs via Reinforcement Learning

Joykirat Singh, Raghav Magazine, Yash Pandya, and Akshay Nambi (Neurips FoRLM Workshop 2025) [\[PDF\]](#)

Exposing the Achilles' Heel: Evaluating LLMs Ability to Handle Mistakes in Mathematical Reasoning

Joykirat Singh, Akshay Nambi, Vibhav Vineet (ACL Main 2025) [\[PDF\]](#)

PROMPTWIZARD: Task-Aware Prompt optimization framework

Eshaan Agarwal, Joykirat Singh, Vivek Dani, Raghav Magazine, T. Ganu, Akshay Nambi (ACL Finding 2025) [\[PDF\]](#)

Data-scarce Behavior Editing of Language Models

Joykirat Singh, Subhabrata Dutta*, T. Chakraborty - (EMNLP Findings 2025) [\[PDF\]](#)*

How to think step-by-step: A mechanistic understanding of chain-of-thought reasoning

Subhabrata Dutta, Joykirat Singh*, S. Chakrabarti, T. Chakraborty - (TMLR 2024) [\[PDF\]](#)*

Frugal LMs Trained to Invoke Symbolic Solvers Achieve Parameter-Efficient Arithmetic Reasoning

Subhabrata Dutta, Joykirat Singh*, I. Pandey*, S. Manchanda, S. Chakrabarti, T. Chakraborty (AAAI 2024) [\[PDF\]](#)*

EROS: Entity-Driven Controlled Policy Document Summarization

Joykirat Singh, Sehban Fazili*, Rohan Jain, Md. Shad Akhtar - (LREC-COLING 2024) [\[PDF\]](#)*

EXPERIENCE

MURGe-Lab Group

Research Assistant (Advisor: Mohit Bansal)

Chapel Hill, NC

Aug 2025 – Present

- Improved the reasoning abilities of language models while enhancing their efficiency through adaptive difficulty-based attention compression. (Under Review ICLR2026)

Microsoft Research

Research Fellow (Principal Researcher: Akshay Nambi)

Bengaluru, India

Jul 2024 – May 2025

- Enhanced large language models' capabilities through agentic RL-based tool integration. (Neurips FoRLM Workshop 2025)
- Investigated LLMs' capacity to detect and correct reasoning errors, revealing fundamental weaknesses in their step-by-step reasoning processes (ACL Main 2025)
- Developed an automated, feedback-driven framework for task-aware and efficient prompt optimization. (ACL Finding 2025)
- Enhanced small language models' mathematical reasoning through dense-reward MCTS training and Direct Policy Optimization for step-wise improvement. (Under Review AAAI 2026)

Laboratory for Computational Social Systems (LCS2), IIT Delhi

Graduate Research Assistant under Prof Tanmoy Chakraborty and Prof Soumen Chakrabarti

New Delhi, India

Jun 2023 – Jun 2024

- Revealed internal mechanisms driving step-by-step reasoning through mechanistic interpretability. (TMLR 2024)

- Improved arithmetic reasoning via symbolic solvers, low-rank adaptation, and RL. (AAAI 2024)
- Developed a data-efficient task adaptation method that aligns model behavior through targeted neural rotations. (EMNLP Finding 2025)

Expedia Group

New Delhi, India

Software Development Engineer, Payment's Module

Aug 2023 – Jun 2024

- Contributed to payment and conversation modules using Java/Kotlin, implementing CI pipelines, Datadog metrics, and LanguageIO-based translation for production systems.

Google Summer of Code

Remote

Intern, MusicBlock

May - Aug 2021

- Contributed to MusicBlock 2.0 using React, TypeScript, and P5, implementing MVVM-based palette and artboard features and improving UX from version 1.0.

HONOR & AWARDS

- Awarded a silver medal for academic excellence at IIITD for achieving highest CGPA in computer science branch
- Awarded the Chanakya Fellowship by iHub Anubhuti IIITD (2022) to work on an independent research problem
- Presented the paper (top ~2%, Oral Presentation), Frugal LMs Trained to Invoke Symbolic Solvers Achieve Parameter-Efficient Arithmetic Reasoning at AAAI'24 Conference
- Volunteer, AAAI'24 Conference - received a travel grant to attend the conference

PROFESSIONAL SERVICES

Conference Reviewer

- ACL 2025, EMNLP 2025 (ACL Rolling Review)
- ICLR 2024-25
- AAAI 2026

REFERENCES

- [Mohit Bansal](#), John R. Louise S. Parker Professor of CS, UNC Chapel Hill.
- [Akshay Nambi](#), Principal Researcher, Microsoft Research India.
- [Tanmoy Chakraborty](#), Associate Professor, IIT Delhi, India.
- [Soumen Chakrabarti](#), Professor, IIT Bombay, India.